

Logan Grout

Cornell University
Operations Research and Information Engineering

Frank H.T. Rhodes Hall
136 Hoy Rd, Ithaca, NY, 14850
lcg58@cornell.edu

Education

PhD Student at Cornell University Operations Research and Information Engineering

2020 MMath University of Waterloo Combinatorics and Optimization

Thesis: *Augmenting Trees to Achieve 2-Node-Connectivity*

Advisor: *Joseph Cheriyan*

2019 BMath University of Waterloo Honours Combinatorics and Optimization, Co-operative Program
Joint Honours Computer Science, Co-operative Program

Journal Publications

Logan Grout, Benjamin Moore. *The Pseudoforest Analogue for the Strong Nine Dragon Tree Conjecture is True*. <https://arxiv.org/abs/1905.02600>. (2019), 10 pages, submitted for journal publication.

Conference Publications

Sylvia Boyd, Joseph Cheriyan, Robert Cummings, Logan Grout, Sharat Irahimpur, Zoltán Szigeti, Lu Wang. *A 4/3-Approximation Algorithm for the Minimum -Edge Connected Multisubgraph Problem in the Half-Integral Case*. <https://arxiv.org/abs/2008.03327>. In Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques *APPROX/RANDOM* 2020.

Awards

2020 NSERC Alexander Graham Bell Canada Graduate Scholarship - Masters
2020 President's Graduate Scholarship
2019 President's Research Award
2017 NSERC Undergraduate Student Research Award
2017 President's Research Award
2015 Ruth Taylor Scholarship (Entrance Scholarship for Outstanding Undergraduate in Mathematics)
2015 University of Waterloo President's Scholarship of Distinction

Teaching

2020 Certificate of Completion of Fundamentals of University Teaching program,
University of Waterloo, Centre for Teaching Excellence

2019 University of Waterloo Teaching Assistant Introduction to Optimization
Applied Cryptography

2017 University of Waterloo Undergraduate Tutor Linear Algebra I

2016 University of Waterloo Undergraduate Grader Introduction to Combinatorics

2015 University of Waterloo Undergraduate Tutor Calculus I

Other Experience

2019	NCC Group	Junior Security Analyst
2018	University of Waterloo (Supervisor: Joseph Cheriyan)	Undergraduate Student Research Assistant Thin-trees, arboricity, and The Nine Dragon Tree Conjecture
	Communications Security Establishment	Junior Crypt-analyst
2017	University of Waterloo (Supervisor: Joseph Cheriyan)	Undergraduate Student Research Assistant Graph Orientations
2016	Communications Security Establishment Digiflare Inc.	Junior Crypt-analyst Junior Quality Assurance Analyst

Talks Given

January 10, 2020	Combinatorial Optimization Reading Group, University of Waterloo Beating $3/2$ for Approximating TSP in the Half-Integral Case
October 10, 2019	Graphs and Matroids Seminar, University of Waterloo A Pseudoforest Analogue of The Strong Nine Dragon Tree Conjecture
July 28, 2019	Canadian Undergraduate Mathematics Conference, Queen's University The Strong Nine Dragon Tree Conjecture